



# SOFTWARE MAINTENANCE

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# INTRODUCTION

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- ▶ The software product must change or evolve. Once in operation, defects are uncovered, operating environments change, and new user requirements surface
- ▶ It has not, historically, received the same degree of attention that the other phases have
- ▶ Post-delivery activities include software modification, training, and operating or interfacing to a help desk.
- ▶ The Software Maintenance KA is related to all other aspects of software engineering





# **Software Maintenance Fundamentals**



# Definitions and Terminology

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- ▶ IEEE Standard for Software Maintenance, IEEE 1219, as the modification of a software product after delivery to correct faults, to improve performance or other attributes, or to adapt the product to a modified environment
- ▶ The IEEE/EIA 12207 describes maintenance as the process of a software product undergoing “modification to code and associated documentation due to a problem or the need for improvement. The objective is to modify the existing software product while preserving its integrity.”





# Nature of Maintenance

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- ▶ Modification requests are logged and tracked, the impact of proposed changes is determined, code and other software artifacts are modified, testing is conducted, and a new version of the software product is released. Also, training and daily support are provided to users.
- ▶ IEEE/EIA 12207 identifies the primary activities of software maintenance as: process implementation; problem and modification analysis; modification implementation; maintenance review/acceptance; migration; and retirement.
- ▶ Maintenance must take the products of the development, code, or documentation, for example, and support them immediately and evolve/maintain them progressively over the software life cycle





# Need for Maintenance

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- ▶ Maintenance is needed to ensure that the software continues to satisfy user requirements
- ▶ Maintenance must be performed in order to:
  - ▶ Correct faults
  - ▶ Improve the design
  - ▶ Implement enhancements
  - ▶ Interface with other systems
  - ▶ Adapt programs so that different hardware, software, system features, and telecommunications facilities can be used
  - ▶ Migrate legacy software
  - ▶ Retire software
- ▶ The maintainer's activities comprise four key characteristics, according to Pfleeger [Pfl01]:
  - ▶ Maintaining control over the software's day-to-day functions
  - ▶ Maintaining control over software modification
  - ▶ Perfecting existing functions
  - ▶ Preventing software performance from degrading to unacceptable levels





# Majority of Maintenance Costs

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- ▶ Studies and surveys over the years have indicated that the majority, over 80%, of the software maintenance effort is used for non-corrective actions
- ▶ Pfleeger [Pfl01] presents some of the technical and non-technical factors affecting software maintenance costs, as follows:
  - ▶ Application type
  - ▶ Software novelty
  - ▶ Software maintenance staff availability
  - ▶ Software life span
  - ▶ Hardware characteristics
  - ▶ Quality of software design, construction, documentation and testing







# Categories of Maintenance

- ▶ Standard for Software Engineering-Software Maintenance, ISO/IEC 14764 to include four categories, as follows:
  - ▶ Corrective maintenance: Reactive modification of a software product performed after delivery to correct discovered problems
  - ▶ Adaptive maintenance: Modification of a software product performed after delivery to keep a software product usable in a changed or changing environment
  - ▶ Perfective maintenance: Modification of a software product after delivery to improve performance or maintainability
  - ▶ Preventive maintenance: Modification of a software product after delivery to detect and correct latent faults in the software product before they become effective faults

|           | <b>Correction</b> | <b>Enhancement</b> |
|-----------|-------------------|--------------------|
| Proactive | Preventive        | Perfective         |
| Reactive  | Corrective        | Adaptive           |



# **Key Issues in Software Maintenance**



# Technical Issues

- ▶ Limited understanding: Limited understanding refers to how quickly a software engineer can understand where to make a change or a correction in software which this individual did not develop
- ▶ Testing: The cost of repeating full testing on a major piece of software can be significant in terms of time and money
- ▶ Impact analysis: describes how to conduct, cost effectively, a complete analysis of the impact of a change in existing software. Maintainers must possess an intimate knowledge of the software's structure and content. States that the objectives of impact analysis are:
  - ▶ Determination of the scope of a change in order to plan and implement work
  - ▶ Development of accurate estimates of resources needed to perform the work
  - ▶ Analysis of the cost/benefits of the requested change
  - ▶ Communication to others of the complexity of a given change





# Management Issues

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- ▶ Alignment with organizational objectives: Organizational objectives describe how to demonstrate the return on investment of software maintenance activities
- ▶ Staffing: refers to how to attract and keep software maintenance staff. Maintenance is often not viewed as glamorous work.
- ▶ Process: Software process is a set of activities, methods, practices, and transformations which people use to develop and maintain software and the associated products
- ▶ Organizational aspects of maintenance: describe how to identify which organization and/or function will be responsible for the maintenance of software. The team that develops the software is not necessarily assigned to maintain the software once it is operational
- ▶ Outsourcing: is becoming a major industry. Large corporations are outsourcing entire portfolios of software systems, including software maintenance





# Maintenance Cost Estimation

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- ▶ Cost estimation: that impact analysis identifies all systems and software products affected by a software change request and develops an estimate of the resources needed to accomplish that change.
- ▶ Experience: Clearly the best approach to maintenance estimation is to combine empirical data and experience. These data should be provided as a result of a measurement program.





# Software Maintenance Measurement

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- ▶ **Specific Measures:** The maintainer must determine which measures are appropriate for the organization in question. [IEEE1219-98; ISO9126-01; Sta94] suggests measures which are more specific to software maintenance measurement programs.
- ▶ That list includes a number of measures for each of the four sub-characteristics of maintainability:
  - ▶ *Analyzability*
  - ▶ *Changeability*
  - ▶ *Stability*
  - ▶ *Testability*





# **Maintenance Process**



# Maintenance Processes

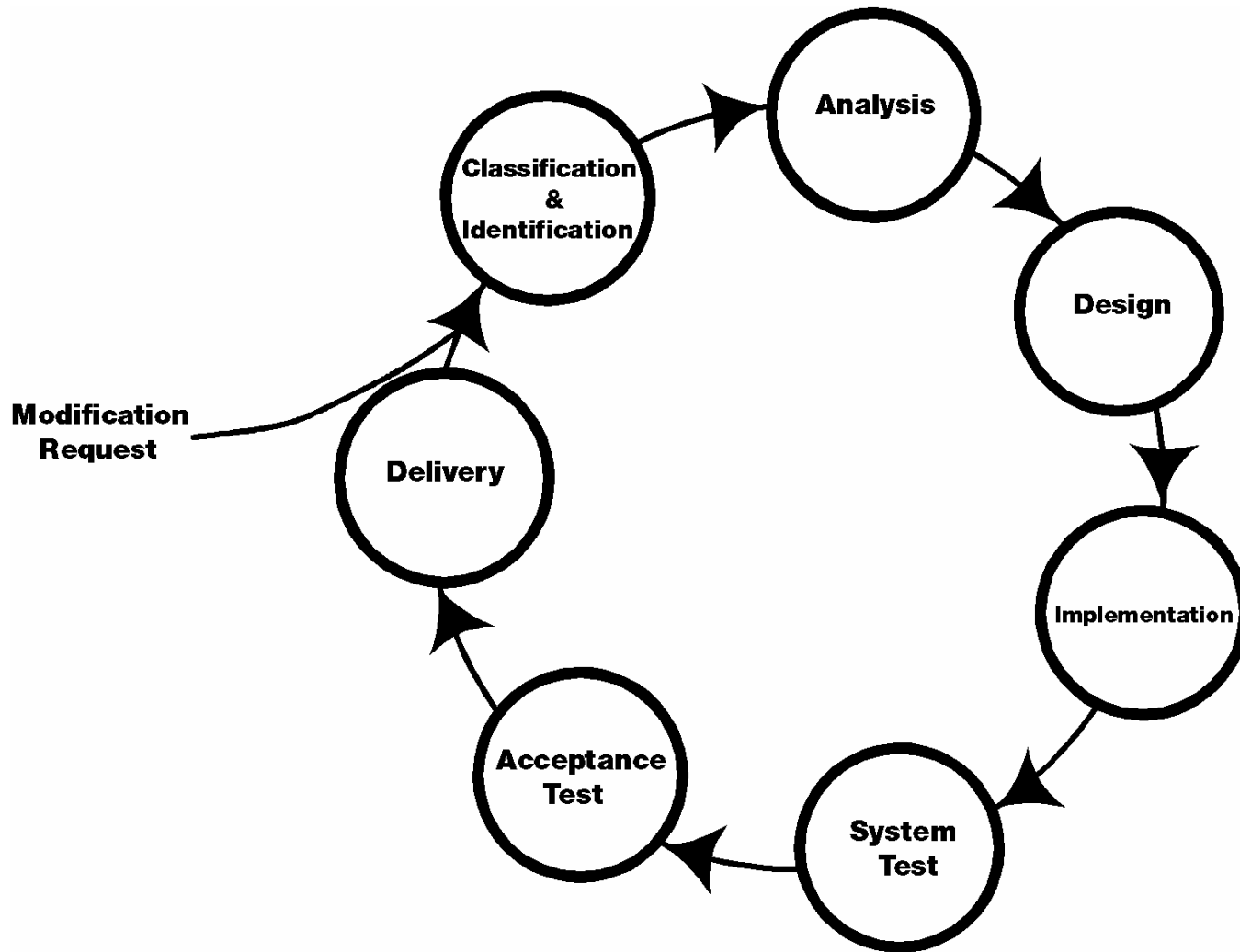
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- ▶ Maintenance processes provide needed activities and detailed inputs/outputs to those activities
- ▶ The maintenance process model described in the Standard for Software Maintenance (IEEE1219) starts with the software maintenance effort during the post-delivery stage and discusses items such as planning for maintenance.





# The IEEE1219-98 Maintenance Process Activities





# Maintenance Activities

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- ▶ Unique activities: There are a number of processes, activities, and practices that are unique to software maintenance, for example:
  - ▶ Transition: is transferred progressively from the developer to the maintainer [Dek92, Pig97]
  - ▶ Modification Request Acceptance/Rejection: size/effort/complexity may be rejected
  - ▶ Modification Request and Problem Report Help Desk
  - ▶ Impact Analysis
  - ▶ Software Support
  - ▶ Service Level Agreements (SLAs)





# Maintenance Activities

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- ▶ Supporting activities: Maintainers may also perform supporting activities, such as software maintenance planning, software configuration management, verification and validation, software quality assurance, reviews, audits, and user training.
- ▶ Maintenance planning activity: An important activity for software maintenance is planning, and maintainers must address the issues associated with a number of planning perspectives:
  - ▶ Business planning (organizational level)
  - ▶ Maintenance planning (transition level)
  - ▶ Release/version planning (software level)
  - ▶ Individual software change request planning (request level)





# Maintenance Activities

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- ▶ Maintenance planning activity:

The release/version planning activity requires that the maintainer [IT101]:

- ▶ Collect the dates of availability of individual requests
  - ▶ Agree with users on the content of subsequent releases/versions
  - ▶ Identify potential conflicts and develop alternatives
  - ▶ Assess the risk of a given release and develop a back-out plan in case problems should arise
  - ▶ Inform all the stakeholders
- ▶ A concept document should be developed, followed by a maintenance plan: The scope of the software maintenance, Adaptation of the process, Identification of the software maintenance organization, An estimate of costs





# Maintenance Activities

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- ▶ Software configuration management: Software configuration management procedures should provide for the verification, validation, and audit of each step required to identify, authorize, implement, and release the software product.
- ▶ The software product and any changes made to it must be controlled. This control is established by implementing and enforcing an approved software configuration manageme
- ▶ Maintainers participate in Configuration Control Boards to determine
- ▶ the content of the next release/version.nt (SCM) process.





# Maintenance Activities

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- ▶ **Software quality:** It is not sufficient, either, to simply hope that increased quality will result from the maintenance of software. It must be planned and processes implemented to support the maintenance process. The activities and techniques for Software Quality Assurance (SQA), V&V, reviews, and audits must be selected in concert with all the other processes to achieve the desired level of quality.



# **Techniques for Maintenance**



# Program Comprehension

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- ▶ Programmers spend considerable time in reading and understanding programs in order to implement changes.
- ▶ Clear and concise documentation can aid in program comprehension.







# Re-engineering

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- ▶ Reengineering is defined as the examination and alteration of software to reconstitute it in a new form, and includes the subsequent implementation of the new form.
- ▶ Dorfman and Thayer [Dor02] state that reengineering is the most radical (and expensive) form of alteration.
- ▶ It is often not undertaken to improve maintainability, but to replace aging legacy software. Arnold [Arn92] provides a comprehensive compendium of topics, for example: concepts, tools and techniques, case studies, and risks and benefits associated with reengineering.





# Reverse Engineering

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- ▶ Reverse engineering is the process of analyzing software to identify the software's components and their interrelationships and to create representations of the software in another form or at higher levels of abstraction.
- ▶ Reverse engineering is passive; it does not change the software, or result in new software.
- ▶ One type of reverse engineering is redocumentation. Another type is design recovery [Dor02].
- ▶ Finally, data reverse engineering has gained in importance over the last few years where logical schemas are recovered from physical databases. (Hen01)





# Referencias

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- ▶ Allain Abran, James W Moore, Swebook Guide to Software Engineering Body of Knowledge. 2004,



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